

1 Introduction

The proliferation of generative artificial intelligence (AI) has sparked a global debate about its potential impact on the labor market. This discourse spans utopian predictions of enhanced productivity, dystopian fears of widespread job displacement, and skeptical views that AI will have minimal effects on employment or productivity. Historically, technologies have affected different tasks, occupations, and industries in different ways, replacing work in some, augmenting others, and transforming still others. These heterogeneous effects suggest that there may be “canaries in the coal mine” which are harbingers of more widespread effects of AI.

Recent discussion coincides with rapid improvements in AI capabilities and adoption. From 2023 to 2024, AI systems improved from solving 4.4% of coding problems to 71.7% on SWE-Bench, a widely used benchmark for software engineering (Maslej et al., 2025). AI has also improved in areas like language understanding, subject knowledge, and reasoning. A recent paper found that current systems could match or outperform up to 47 percent of industry professionals on a pre-defined benchmark of economically valuable tasks (Patwardhan et al., 2025). At the same time, AI systems are increasingly widely adopted. Hartley et al. (2025) find that LLM adoption at work among U.S. survey respondents above age 18 reached 46% by June/July 2025.¹

Given the better capabilities and widespread adoption, a central concern, amplified in recent headlines, is whether AI is beginning to supplant human labor, particularly for younger, entry-level workers in highly exposed professions like software engineering and customer service.² For instance, in May of 2025, Dario Amodei, co-founder and CEO of Anthropic predicted that AI could wipe out roughly 50 percent of all entry-level white-collar jobs within five years (Morris, 2025).

Despite the intensity of this debate, empirical evidence has struggled to keep pace with technological advancement, leaving many fundamental questions unanswered. This paper confronts this empirical gap by leveraging a large-scale, high-frequency administrative dataset from ADP, the largest payroll software provider in the United States. Our sample consists of monthly, individual-level payroll records through September 2025, encompassing millions of workers across tens of thousands of firms. This rich panel structure allows us to track employment dynamics with a high

¹Similarly, Bick et al. (2024) found that in late 2024, nearly 40% of the U.S. population age 18-64 reported using generative AI, with 23% using it for work weekly and 9% daily.

²Improved productivity of workers in an occupation could lead to either reduced or increased employment, depending on, among other things, how elastic demand is for the output of those workers.

have a college degree.

Occupations with a high share of college graduates have declining employment overall, with muted differences between more-exposed and less-exposed occupations compared to our main results. In contrast, occupations with a low share of college graduates have rising overall employment, with the least AI-exposed occupations growing and the most exposed occupations declining in employment. Further, for lower college share occupations, the dispersion in employment outcomes is visible in higher age groups as well, with workers up to age 40 showing separation in employment trends by AI exposure. These findings suggest that deteriorating education outcomes cannot fully explain our main results. They also suggest that for non-college workers, experience may serve as less of a buffer to labor market disruption than for college workers.

Occupational Interest Rate Exposure While estimates with firm-time effects control for overall hiring trends within firms, another possibility is that even within firms occupations with high AI exposure have high interest rate exposure, leading to contraction in hiring. We use occupational interest rate exposure data from [Zens et al. \(2020\)](#) to test for the correlation between interest rate exposure and AI exposure. Figure [A22](#) in fact finds that AI exposure and interest rate exposure are negatively correlated overall, with occupations such as construction having high interest rate exposure and low AI exposure. Figures [A23](#) and [A24](#) repeat our analysis separately for occupations below and above the median in interest rate exposure. Both cases are consistent with our main results, suggesting that occupational differences in interest rate exposure do not drive our results.

Other Robustness Checks Figures [A25](#) and [A26](#) show results separately for men and women. The results are similar, suggesting that diverging prospects for men and women are not driving our findings. Figure [A27](#) shows that results are similar when we do not take a balanced sample of firms. Figure [A28](#) shows similar results when including part-time and temporary workers.²⁷

Comparison to CPS Data A useful benchmark for our findings is to compare them to estimates from the monthly Current Population Survey (CPS). In Online Appendix [C](#) we perform similar analyses in the CPS, finding high levels of noise consistent with small sample sizes in fine age-

²⁷Another possibility is that employment trends reflect Covid-19 stimulus checks distorting labor supply. However, these payments were income-conditioned, and AI-exposed occupations average higher incomes ([Kochhar, 2023](#)), making this channel unlikely in explaining our findings.

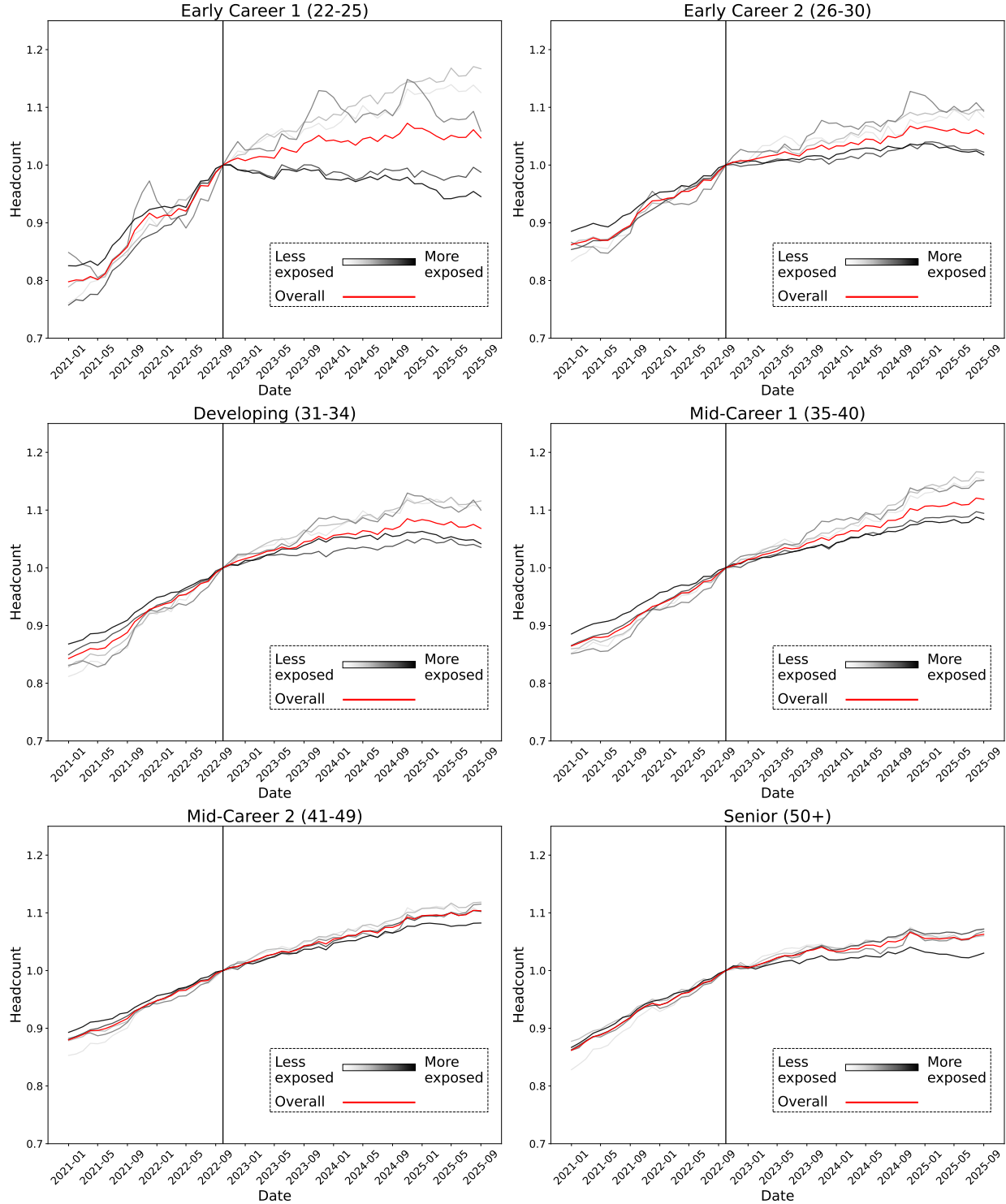


Figure A25: Employment changes by age and exposure quintile using measures from [Eloundou et al. \(2024\)](#). Considering only men.

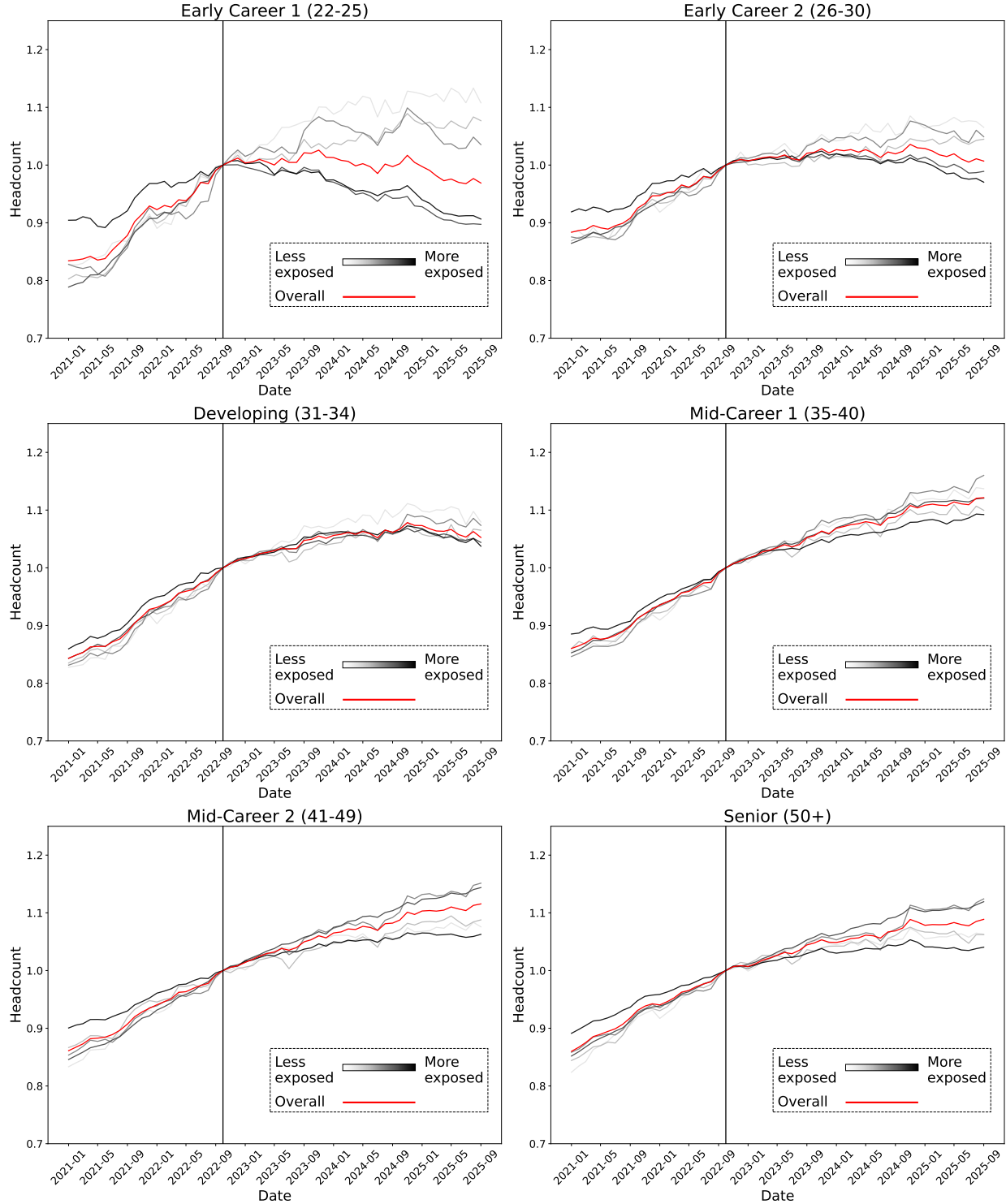


Figure A26: Employment changes by age and exposure quintile using measures from [Eloundou et al. \(2024\)](#). Considering only women.